

Short Answer Test

1. Data Science can analyze past academic records (attendance, assignment scores, internal marks, participation) to build a predictive model that flags at-risk students early. Techniques like logistic regression or decision trees identify patterns correlated with failure, enabling timely intervention before the final exam.
2. Data Science enables predictive analytics on historical patient data - diagnosing disease trends, predicting readmission risk, optimizing treatment plans and identifying outbreak patterns. Data mining and machine learning models can also assist in personalized medicine and resource planning based on patterns in years of records.
3. It is an application of Recommendation Systems, a Data Science technique using collaborative filtering or content-based filtering to suggest relevant products, increasing sales and customer satisfaction.
4. It uses Anomaly / Fraud Detection, where Data Science models learn normal transaction patterns and flag deviations in real time as potentially fraudulent, helping prevent financial loss.

5. It is Customer Churn Prediction. Data Science analyzes customer behavior, purchase history, and engagement data to build models predicting which customers are likely to leave, allowing the company to take proactive retention measures.
6. One benefit: Data Science can identify learning gaps and performance patterns through analysis of test scores and attendance, helping teachers personalize instruction and intervene early for weaker students - improving overall academic outcomes.
7. Supermarket festival sales trends - Data Science applies descriptive and predictive analytics on historical sales data to identify which products sell more during festivals, helping with inventory planning and targeted promotions.
8. Data Science can analyze real-time traffic + sensor data using predictive analytics to forecast congestion patterns, optimize traffic signal timing, and suggest alternate routes - improving overall traffic flow management.
9. Farmer improving crop yield, it use predictive analytics using weather, soil, and crop data to recommend optimal sowing times, irrigation schedules and fertilizer usage helping increase yield and reduce resource wastage.

10. This shows the Variety part of Big Data - data exists in structured, unstructured (reviews, text) and multimedia (images, videos) formats simultaneously.
11. This shows the Velocity part of Big Data - the speed at which data is generated, streamed, and must be processed in real time or near real time.
12. This is Structured Data, organized in a fixed schema (tables with defined rows and columns), typically stored in relational databases.
13. This reflects the Volume characteristic of Big Data - the sheer scale / quantity of data generated continuously by users.
14. This is Unstructured Data - text data with no fixed format or schema, requiring techniques like NLP to process.
15. This indicates a Veracity issue - the data is inconsistent, incomplete or untrustworthy, requiring data cleaning before analysis.
16. This highlights the Value characteristic of Big Data - the ability to extract meaningful actionable insights from raw data that drive business decisions.

17. This uses the Storage layer of the Big Data ecosystem - specifically HDFS, which stores massive datasets across distributed clusters in a fault-tolerant manner.
18. Apache Spark (or Hadoop MapReduce) can be used - these frameworks distribute computation across multiple nodes, enabling fast, parallel processing of large-scale data.
19. This requires a data ingestion/streaming component of the Big Data ecosystem, such as Apache Kafka or Apache Flume, designed to collect and transmit continuous real-time data streams.
20. Data Visualization tools such as Tableau, Power BI, or Python libraries can be used to present analytical results as dashboards, charts, and graphs for easier interpretation by decision makers.